

Final Project Report: Predicting Stock-Level Returns Using Machine Learning

Simrat Kaur Randhawa
Master's in Computer Science
Stevens Institute of Technology

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1. Introduction

This project aims to develop and compare machine learning models for predicting monthly stock-level returns using firm-specific and macroeconomic characteristics. Models are evaluated using both traditional prediction metrics (e.g., R^2 , MSE) and financial performance metrics (e.g., average return, volatility, Sharpe ratio) through the construction of a top-100 stock portfolio.

2. Dataset

The dataset includes over 90 firm-level financial features and macroeconomic variables spanning from 1957 to 2021. The target variable is the monthly stock return.

Preprocessing Steps:

- Median imputation for missing values
- Feature standardization
- Creation of return deciles and feature groups

3. Exploratory Data Analysis (EDA)

- Returns showed high kurtosis and heavy tails.
- Features like `rd_mve`, `ill`, and `retvol` correlated positively with returns.
- Features like `invest`, `hire`, and `chcsho` correlated negatively with returns.

4. Feature Grouping

Thematic categories:

- **Valuation:** `bm`, `ep`, `dy`, `sp`, `agr`
- **Momentum:** `mom1m`, `mom6m`, `mom12m`, `mom36m`

- **Profitability:** roeq, cfp, gma
- **Liquidity:** ill, baspread, turn, retvol
- **Macro:** macro_*

5. Models Implemented

- **Linear Regression:** Baseline model with low R^2 and weak returns.
- **Lasso Regression:** Improved via regularization; Optuna tuning enhanced performance.
- **XGBoost:** Strong in both prediction and portfolio metrics; best Sharpe ratio overall.
- **Neural Network (MLP):** Initially weak, later outperformed others in volatile periods.

6. XGBoost vs. Neural Network

Both XGBoost and Neural Network demonstrated strong performance on the data but in different ways

- **XGBoost:** Consistently delivered strong predictive accuracy and stable returns throughout the dataset, particularly in early periods where return patterns were more structured.
- **Neural Network:** While showing poor R^2 performance, excelled in identifying high return stocks. The cumulative returns in the final months seemed to exceed XGBoost, suggesting superior adaptability in volatile or regime shifting markets.
- **Interpretability:** XGBoost favored due to feature importance and tree structure.
- **Overfitting:** XGBoost more robust; NN required regularization techniques. This comparison emphasizes that while XGBoost is the better general-purpose predictor, NNs may offer an edge during market transitions when feature-return relationships become nonlinear or noisy.
- **Hyperparameter Tuning:** Both XGBoost and Neural Network were tuned using optuna, for Neural Network the returns improved significantly but still XGBoost worked the best.

7. Advanced Techniques

For the pupose of exploration I tried out a few advanced techniques surrounding ensemble learning and residual learning.

- **Feature Group Ensemble:** Combined XGBoost on groups using equal weight and stacking. Mixed results.
- **Residual Learning:** Tested Lasso/XGB + NN and NN + XGB. No consistent improvement over base models.

8. Evaluation Metrics

Model	Avg Return	Volatility	Sharpe	Test R^2
Linear Regression	0.0072	0.0896	0.0809	-0.0388
Lasso	0.0145	0.0546	0.2648	0.0093
Lasso (Optuna)	0.0240	0.1023	0.2349	0.0190
XGBoost	0.0313	0.1110	0.2817	-
Neural Network	0.0321	0.1153	0.2782	~ 0.0000
Ensemble (equal)	0.0180	0.1199	0.1501	-
Stacked Ensemble	0.0142	0.1312	0.1081	0.0065
Hybrid (XGB+NN)	0.0269	0.1146	0.2344	-

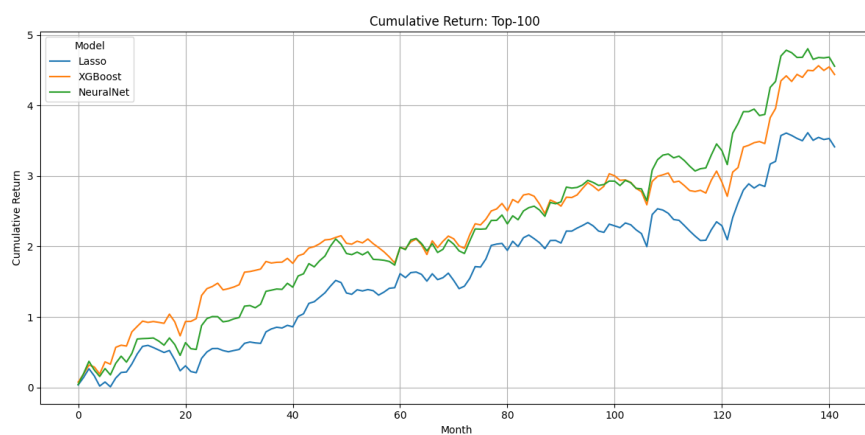


Figure 1: Top 100 Stock Returns

9. Key Observations

- XGBoost and Neural Networks were the top performers.
- NN slightly outperformed XGBoost during the final months on the returns front but looking at overall performance XGBoost comes out victorious.
- Residual learning and stacking did not offer significant benefits.

10. Conclusion

While ensemble and hybrid models added complexity, they did not outperform individual strong models. Neural Networks showed adaptability under changing conditions, whereas XGBoost remained a robust, interpretable baseline. Looking at overall performance XGBoost seems to be the state of the art for this dataset.

11. Some other things I wanted to try but was unable to due to time and resource constraints

- Explore classification-based portfolio selection

- Implement quantile regression for tail risk modeling
- Use rolling-window retraining for dynamic adaptation
- Visualize feature drift using SHAP or permutation importance

Note

ChatGPT was used in writing this report and while debugging a few dependancy issues.